Cover letter: European Journal of Nuclear Medicine and Molecular Imaging

Dear Editors,

Please consider my manuscript, "Oxygenation versus tissue composition in positronium-lifetime PET: an identifiability limit for hypoxia imaging", for publication as an Original Article in the *European Journal of Nuclear Medicine and Molecular Imaging*.

Positronium lifetime imaging is moving into in-vivo long-axial-field-of-view PET/CT, and tissue hypoxia is one of its most clinically attractive proposed applications. The central question addressed here is whether a positronium-lifetime contrast can be interpreted as oxygenation in biological tissue, where composition and free-volume effects also change the lifetime. We show that from the lifetime alone, oxygenation and composition are structurally non-identifiable: both enter one summed decay rate. We then quantify the resulting apparent-pO2 bias, derive CRLB limits for a second-observable rescue using o-Ps intensity, and translate the limits into count and volume requirements for realistic PET scenarios.

The manuscript is intended as a field-level quantitative boundary for an emerging nuclear-medicine biomarker. Its main result is constructive rather than dismissive: lifetime-only maps should not be interpreted as hypoxia, but the problem becomes identifiable in principle when a second observable such as o-Ps intensity or a 3gamma/2gamma ratio is measured. The revised submission also includes sensitivity analyses for the two load-bearing quantities–the composition dependence of I3 and the unmeasured lipid quenching constant–and specifies the bench PALS experiment needed to measure them directly.

This is a modelling and methods study using no human or animal data. The work is not under consideration elsewhere. All code and generated outputs are available in the public repository listed in the manuscript.

Sincerely,

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